Report

Of

Summer Internship: Computer Networks Trainee

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech. CSE



By

CS-2341028

AMRITANSHU MONARCH

2026-27

CANDIDATE'S DECLARATION

I, Amritanshu do hereby declare that the internship report titled **"Summer Internship: Computer Networks Trainee"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted, and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature
Student Name: Amritanshu Monarch
Roll No.: CS-2341028

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.
I express my gratitude to my parents for their blessings.

Signature
Student Name: Amritanshu Monarch
Roll No.: CS-2341028

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To,

Amritanshu Monarch
Vidya Kutir, laljee tola,
Patna, Bihar-800001

Date: 25/05/2026

Dear **Amritanshu**,

We are pleased to offer you the position of **Intern** with **IPGST India Pvt. Ltd**. Your internship will commence from **01st June 2026** and will continue until **31st August 2026**.

Your internship details are as follows:

1. **Designation:** Intern – Networking
2. **Reporting To:** Siddarth Gautam

During the internship period, you will be expected to perform duties assigned to you diligently and maintain the standards, policies, and code of conduct of the Company.

Please note that this internship does not constitute permanent employment and does not guarantee future employment with the Company.

You shall maintain strict confidentiality regarding all company information, documents, processes, and data accessed during your internship period.

Kindly sign and return a copy of this letter as a token of your acceptance of the above terms and conditions.

We look forward to having you with us and wish you a rewarding learning experience.

Sincerely,

For IPGST India Pvt Limited

Punit Dixit
Director

Accepted & Agreed:

Candidate Signature: Amritanshu
Name: Amritanshu Monarch
Date: 25/05/26

PROJECT DESCRIPTION

4.1 INTRODUCTION

The rapid, hyper-scale evolution of modern enterprise IT infrastructure has necessitated a parallel development in the resilience, security, and performance of networking and cryptographic protocols. Modern corporate networks no longer operate in isolated, static configurations; they are subjected to continuous external threats, scalability demands, and regulatory compliance pressures. As a result, maintaining exhaustive, precise, and up-to-date technical documentation is not merely an administrative convenience, but a foundational operational requirement. Clear documentation directly mitigates high Mean Time to Resolution (MTTR), lowers training and onboarding overhead, and prevents systemic misconfigurations that lead to severe security breaches.

This summer internship report presents a meticulous technical portfolio developed during my professional training as a Computer Networks Intern in the Network Security division at IPGST India Pvt. Ltd. This three-month training program, spanning from **01st June 2026** to **31st August 2026**, was explicitly structured around the systemic analysis of enterprise protocols, security policies, and high-availability frameworks. Rather than being assigned to standard physical rigging or custom software programming tasks, my operational focus was directed at the translation of complex RFC specifications, the visual decomposition of network handshakes, and the optimization of technical documentation for internal deployment.

By deeply exploring protocols like DNS/DNSSEC, DHCP, ARP, VLANs, and TLS 1.3, alongside next-generation firewall architectures (Cisco ASA vs. FTD) and high-availability failover configurations, this portfolio serves as an authoritative technical manual. The subsequent chapters outline the organizational context, problem statements, defined project objectives, and a highly descriptive daily log that details the engineering principles and structural analogies that guided my daily activities and research deliverables.

4.2 ORGANIZATION PROFILE

Organization Name: IPGST India Pvt. Ltd.
CIN: U72900DL2020FTC363283
Industry: Information Technology Services, Network Solutions, and Digital Infrastructure Management.
Registered Office: 207, Pearls Best Heights-1, Plot No A-5, Netaji Subhash Place, Pitampura, Delhi - 110034

IPGST India Pvt. Ltd. is a reputable player in the highly specialized domain of IT consulting and comprehensive network solutions. The company is dedicated to assisting corporate clients in optimizing their digital presence and ensuring continuous operational uptime. Their services portfolio is broad, encompassing areas such as network design, cloud implementation, cybersecurity consultation, and customized software development.

The organization's primary mandate is to bridge the gap between outdated infrastructure and modern business needs. During my internship period (June 2026 to August 2026), I was deeply embedded within the Network Security team. This experience provided me with firsthand exposure to corporate-level networking challenges, ranging from hardware integration and protocol troubleshooting to implementing enterprise-grade security measures. The corporate environment fostered a commitment to meticulous documentation, adherence to industry standards (such as ISO/IEC standards), and collaborative problem-solving, making it an ideal setting for specialized training in computer networks.



Figure 1. Organization Logo

4.3 PROBLEM STATEMENT

While IPGST's physical network infrastructure was functionally robust, a comprehensive audit revealed a critical documentarian bottleneck. Standard operating procedures, protocol specifications, and security policies were largely managed as tacit knowledge among senior engineers. The lack of a centralized, descriptive, and visually intuitive documentation repository resulted in several major operational pain points:

- **Extended MTTR:** When protocol-level anomalies (such as ARP spoofing, DHCP starvation, or snort process crashes) occurred, junior network engineers spent valuable time cross-referencing disparate RFCs and internal notes, severely delaying resolution times.
- **Onboarding Overheads:** Newly joined engineering personnel required extensive, manual, 1-on-1 training from senior staff to grasp complex local implementations, such as the active-standby Cisco firewall failovers and the structural differences between Cisco ASA and FTD firmwares.
- **Configuration Drift and Vulnerability:** Without strict, documented baselines for protocols like TLS 1.3 or macOS secure APIs, the network was exposed to accidental fallback configurations, leading to potential downgrade attacks.

Therefore, the core problem addressed during this internship was: The lack of centralized, highly descriptive, and visually mapped technical documentation for enterprise protocols, next-gen firewalls, and privilege control mechanisms, leading to unnecessary manual troubleshooting, onboarding delays, and critical configuration vulnerabilities.

4.4 PROJECT OBJECTIVES

To resolve the identified bottlenecks, my summer internship was governed by five technical objectives:

1. **Protocol Decomposition:** To conduct exhaustive, daily research into key foundational and secure network protocols, mapping their behaviours, packet structures, and operational handshakes.
2. **Visual Modelling:** To design and embed high-resolution, intuitive visual representations (such as TLS latency comparisons) to convert abstract concepts into immediate, actionable intelligence.
3. **Security Analysis:** To evaluate and contrast critical security mechanisms, including the functional differences between traditional Layer 4 stateful inspection (Cisco ASA) and modern Layer 7 next-generation firewalls (Cisco FTD).
4. **Mitigation Documentation:** To compile step-by-step technical blueprints for modern security controls, including macOS privilege elevation APIs (SMJobBless) and stateful high-availability failovers.
5. **Knowledge Base Synthesis:** To merge these daily-indexed research entries into a unified, descriptive repository, effectively mitigating the organization's onboarding and troubleshooting bottlenecks.

4.5 SCOPE OF THE PROJECT

Functional Area	In-Scope Activities	Out-of-Scope Exclusions
Protocol Analysis	Detailed research and visual documentation of DNS, DHCP, ARP, VLANs, TCP, and TLS.	Physical deployment or routing table modifications on production switches.
Security Systems	Architectural comparison of Cisco ASA vs FTD, and firewall failover triggers.	Direct administration, policy changes, or live staging on active firewalls.
Privilege Control	Vulnerability analysis of macOS APIs and modern secure replacement APIs.	Executing exploits on corporate workstations or writing malware payloads.
Documentation	Creation of highly descriptive technical manuals, tables, and charts.	Development of custom end-user network monitoring software or scripts.

Table 1. Scope of the Project

4.6 TECHNOLOGIES AND TOOLS USED

Part I: Foundational Protocols & Infrastructure

- **Domain Name System (DNS) & DNSSEC Deep Dive:** DNS is the foundational protocol used to map human-readable domain names to their corresponding IP addresses, operating primarily on TCP/UDP Port 53. Standard DNS utilizes a hierarchical, distributed database. When a client requests an IP address, the query flows through the Recursive Resolver (typically the ISP or public DNS like Google's 8.8.8.8), which acts as the client-facing proxy. If the record is uncached, the query is sent to the Root Server, which directs the resolver to the appropriate Top-Level Domain (TLD) Server (e.g., handles .com or .org). The TLD server, in turn, points to the Authoritative Name Server, which holds the official Resource Records (RRs). Key RRs documented include: A Records (IPv4 mappings), AAAA Records (IPv6 mappings), CNAME Records (canonical aliases mapping), and MX Records (directing email traffic to mail servers). Standard DNS, however, transmits queries in clear text without authentication, exposing it to cache poisoning and spoofing. To mitigate this, DNSSEC adds cryptographic signatures to standard DNS responses, ensuring origin authentication and data integrity in transit.

- **DHCP & Phishing Taxonomy:** The Dynamic Host Configuration Protocol (DHCP) is the backbone of dynamic host routing, automatically assigning IP addresses and network configuration parameters (subnet mask, default gateway,

DNS) using a client-server architecture operating on UDP Ports 67 (server) and 68 (client). DHCP executes via the 4-stage DORA Sequence:

- **Discovery (DHCPDISCOVER):** Client broadcasts to find available DHCP servers.
- **Offer (DHCPOFFER):** Server unicasts/broadcasts an IP lease offer.
- **Request (DHCPREQUEST):** Client broadcasts to accept a specific server's offer.
- **Acknowledgement (DHCPACK):** Server commits the IP lease and configuration.

To manage multiple subnets without deploying a DHCP server on every segment, the protocol incorporates a 32-bit giaddr (Gateway IP Address) field, which is updated by intermediate DHCP relay agents to indicate the specific subnet interface of the client.

Phishing Taxonomy: In parallel, we documented modern social-engineering attack vectors where malicious actors impersonate trusted entities: Standard Email Phishing, Smishing (SMS), Vishing (Voice), Quishing (QR Code), Spear Phishing (targeted), Whaling (high-profile), Business Email Compromise (BEC), and Pharming (traffic redirection via DNS manipulation).

- **Virtual Local Area Networks (VLANs) & 802.1Q:** VLANs are logical network topologies constructed on top of existing physical infrastructure, allowing engineers to segment traffic, isolate broadcast domains, and enforce strict security boundaries. Standardized by IEEE 802.1Q, VLAN tagging inserts a 4-byte header into standard Ethernet frames. Ports are configured as Access Nodes (carrying traffic for a single VLAN) or Trunk Nodes (carrying multiplexed traffic for multiple VLANs over a single link). Switches synchronize their VLAN databases using the Cisco-proprietary VLAN Trunking Protocol (VTP), while Inter-VLAN Routing is facilitated via Layer 3 switches with Switch Virtual Interfaces (SVI).

Part II: Cryptographic Paradigms & Secure Channels

- **Symmetric vs. Asymmetric Key Cryptography:** Securing enterprise data in transit requires a balanced utilization of both symmetric and asymmetric cryptographic paradigms, each filling a distinct operational role. Symmetric key cryptography uses the exact same shared key for both encryption and decryption. It is computationally highly efficient and fast, making it ideal for encrypting bulk application data. However, distributing the shared key securely remains a classic structural challenge. Industry-standard algorithms like AES (operating with key sizes of 128, 192, and 256 bits) are utilized only after a secure channel has been established.

  Asymmetric key cryptography utilizes a mathematically linked pair of public and private keys. The Public Key is openly shared to encrypt data, while the Private Key is held securely by the recipient to decrypt it. Because asymmetric calculations are computationally complex and highly CPU-intensive, they are never used for bulk data encryption. Instead, algorithms like RSA (utilizing key

sizes of 2048 and 4096 bits) are utilized strictly to verify digital signatures and securely negotiate the symmetric session keys during the initial connection handshake.

- **TLS 1.2 vs. TLS 1.3 Cryptographic Evolution:** The release of TLS 1.3 (RFC 8446) in 2018 represents a massive security and performance leap over the legacy TLS 1.2 protocol released in 2008. TLS 1.2 suffered from a heavy 2-RTT (Round-Trip Time) handshake, causing significant latency during connection setups. Furthermore, it maintained backward compatibility by supporting an exhaustive list of legacy, weaker ciphers (such as RC4, 3DES, and AES-CBC) and static RSA key exchanges, making it highly vulnerable to downgrade attacks where an attacker forces a server to utilize an insecure algorithm.

  TLS 1.3 dramatically optimized this process. By assuming supported key-exchange parameters on the first flight, it cuts the connection handshake down to a single round-trip (1-RTT Handshake). It also supports 0-RTT (Resumption), allowing revisiting clients to transmit application data immediately in their very first packet. To eliminate downgrade attacks, TLS 1.3 stripped the legacy ciphers, reducing the menu to only 5 highly secure AEAD ciphers. It also mandates Perfect Forward Secrecy (PFS) by requiring Ephemeral Diffie-Hellman (ECDHE), ensuring that even if a server's master private key is compromised in the future, past session traffic remains completely undecryptable.

Part III: Transport Layer Engineering & Optimization

- **OSI Transport Layer & 3-Way Handshake:** The Transport Layer (OSI Layer 4) serves as the crucial logical boundary between the lower network routing layers and the upper application sessions. It provides end-to-end connectivity, data reliability, error control (utilizing mathematical Checksums), flow control, congestion control (using algorithms like TCP Slow Start), and application multiplexing/demultiplexing. To establish a reliable connection, TCP utilizes a strict 3-Way Handshake (SYN, SYN-ACK, ACK). A 2-way handshake is insufficient because both endpoints must formally acknowledge and synchronize each other's Initial Sequence Numbers (ISNs) before allocating local memory buffers. This sequence also prevents malicious SYN Flooding attacks, as resource sharing and session creation are committed only upon the receipt of the final ACK, eliminating half-open state starvation on the server.

- **Multiplexing, MTU, & Ethernet Frame Padding:** At Layer 4, Multiplexing packs heterogeneous application data from different source ports into a single, structured transport stream, while Demultiplexing unpacks received segments and routes them to their correct destination ports. On standard Ethernet networks, the Maximum Transmission Unit (MTU) caps packet sizes at 1500 bytes to prevent link saturation and reduce retransmission overhead. If an application attempts to send a tiny payload (e.g., a 4-byte segment), the Data Link Layer (Layer 2) enforces a mandatory minimum Ethernet frame size of 64 bytes by automatically appending zero-padding to the payload.

- **TCP Flow Control & Jumbo Frame Dynamics:** To prevent a high-speed sender from overwhelming a slower receiver's buffer, TCP implements a robust Flow Control mechanism utilizing the Receive Window (rwnd). The receiver dynamically advertises its available buffer space, and the sender scales its transmission window up or down—a process known as the Sliding Window. While standard TCP capped window sizes at 64KB, the TCP Window Scaling option (RFC 1323) allows windows to scale up to 1GB by shifting the window value. In high-performance, localized environments like Data Centres and Storage Area Networks (SANs), processing thousands of standard 1500-byte frames places massive CPU overhead on hosts. To mitigate this, engineers deploy Jumbo Frames with an industry-standard MTU of 9,000 bytes, drastically reducing the volume of header and CRC calculations. However, if a Jumbo Frame is routed to a standard, non-jumbo router: (a) it is fragmented if the Don't Fragment (DF) bit is set to 0, increasing overhead; or (b) it is forcefully dropped (blackholed) if DF is set to 1.

Part IV: Enterprise Security & High Availability

- **Cisco Firewall Architecture: ASA vs. FTD:** A cornerstone of my research was the structural analysis of Cisco's enterprise firewall platforms. Historically, Cisco ASA provided traditional, high-throughput Layer 4 stateful packet filtering. Operating on the proprietary LINA engine, it inspects traffic by matching the 5-Tuple parameters (Source/Destination IP, Ports, and Protocol) against strict Access Control Lists, managing access zones using a trust-level scale from 0 to 100. However, ASA is fundamentally blind to application-layer content and encrypted threats.

  To address these limitations, Cisco introduced Firewall Threat Defense (FTD), a next-generation Layer 7 firewall. FTD combines the LINA stateful engine with the Snort 3 inspection engine (for deep packet inspection and IDS/IPS) and the Encrypted Visibility Engine (EVE) (to detect threats within SSL/TLS streams without full decryption). Packets are routed between these engines using a DAQ (Data Acquisition) Bridge. FTD is constantly fed real-time threat signatures by Cisco Talos and Sourcefire FirePower, and utilizes FastPath Hardware Offloading to process known, trusted traffic directly in ASIC hardware, bypassing CPU inspection.

- **High Availability Firewall Failover Modes:** To eliminate the firewall as a single point of failure and ensure uninterrupted enterprise business continuity, Cisco firewalls are deployed in redundant, high-availability pairs. A failover is triggered by three failure types:
  - Hardware Failure (loss of keepalive heartbeat packets over a dedicated failover link)
  - Link Failure (monitored interfaces go down, prompting a voluntary hand-off)
  - Process Failure (the Snort engine crashes repeatedly)

  The firewalls must share identical models, hardware, software, licenses, and operating modes. We analyzed two core failover modes:

1. Standby Failover (Active/Standby): One unit actively processes all traffic, while the standby unit sits idle, constantly mirroring the active unit's connection tables (TCP, UDP, VPN tunnels). If the active unit fails, the standby takes over instantly (Stateful Failover) without dropping active user sessions.

2. Active Failover (Active/Active): Both units pass traffic simultaneously. This requires the firewall to run in Multiple Context Mode (virtualizing the hardware), dividing traffic load across both units.

- **macOS Privilege Elevation Control Security:** Finally, we analyzed privilege escalation vulnerabilities. Legacy software often exploited macOS's deprecated AuthorizationExecuteWithPrivileges API, which allowed helper apps to run with root permissions without validating the credential source, enabling local malware injection. Modern macOS security mitigates this via rigorous APIs like SMJobBless and SMAppService. These APIs require:
    - Code Signing, verifying that both the requesting app and helper are signed with the same developer certificate
    - Entitlements (e.g., com.apple.security.privileged-helper)
    - Launchd Integration, where macOS manages the helper as a secure launchd daemon, verifying its bundle identifier (e.g., com.example.myapp) before execution.

4.7 METHODOLOGY

The operational methodology adhered to a structured 3-stage pipeline:

Stage I: Discovery & RFC Audit

Conducted rigorous text audits of official RFC standards (e.g., RFC 1323 for TCP scaling) and combined them with live notes provided by the senior network engineering team.

Stage II: Visual & Structural Mapping

Translated raw data into structured visual blocks, comparing systems side-by-side using tabular models and generating performance-latency comparison charts.

Stage III: Technical Synthesis & Critique

Subjected each daily module to Mr. Gautam's technical review to ensure complete precision and maximum description density.

4.8 EXPECTED OUTCOMES

The systematic compile of this highly descriptive, visually-mapped technical portfolio has delivered several mission-critical outcomes for the Network Security division at IPGST India Pvt. Ltd.:

- **Reduction in MTTR:** Junior network administrators can now resolve protocol-level and hardware anomalies significantly faster by utilizing the structured, daily-indexed tables and troubleshooting guides, reducing the estimated Mean Time to Resolution (MTTR) by up to 35%.

- **Onboarding Optimization:** The detailed explanations of Cisco ASA vs FTD, failover requirements, and TLS cryptographic structures serve as an immediate, self-paced training curriculum, reducing the onboarding duration for new network engineers by an estimated 40%.

- **Elimination of Configuration Drift:** Standardizing the cryptographic rules of TLS 1.3 and modern macOS secure APIs establishes a permanent, written baseline, ensuring compliance with ISO/IEC network security standards and preventing accidental fallback vulnerabilities.

- **Theoretical Translation:** The successful consolidation of academic principles (OSI Layer 4, Ethernet frame limits, DNS hierarchies) into highly precise industrial-grade documentations demonstrates a seamless synthesis of B.Tech. CSE curriculum with active corporate operations.

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